

Baseline Adjustment with ANCOVA

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Introduction

Analysis of Covariance (ANCOVA) regresses the outcome on treatment and baseline value jointly. Compared to a naive change-from-baseline t-test, ANCOVA gains precision, handles regression to the mean correctly, and reduces SE roughly by a factor of $\sqrt{1 - \rho^2}$ where ρ is baseline-outcome correlation.

Prerequisites

Linear regression; correlation; change scores.

Theory

Naive change-score analysis: compare $\Delta = Y_{\text{post}} - Y_{\text{baseline}}$ across arms. Unbiased under randomisation but less efficient than ANCOVA when baseline correlates with outcome.

ANCOVA: $Y_{\text{post}} = \alpha + \beta_{\text{trt}} \cdot T + \beta_{\text{base}} \cdot Y_{\text{baseline}} + \varepsilon$. Treatment effect β_{trt} has lower SE than the change-score test.

Regression to the mean: if groups have different baselines by chance, change scores bias toward the difference; ANCOVA corrects for this.

Assumptions

Linear relationship between baseline and outcome; no treatment-by-baseline interaction (common extension: stratify or add interaction term).

R Implementation

```
set.seed(2026)
n <- 200
baseline <- rnorm(n, 10, 2)
arm <- factor(rep(c("ctrl", "trt"), each = n/2))

# Outcome correlated with baseline; true trt effect = 1
outcome <- 0.7 * baseline + ifelse(arm == "trt", 1, 0) +
  rnorm(n, 0, 1)

# Naive change-score analysis
change <- outcome - baseline
t.test(change ~ arm)

# ANCOVA
fit <- lm(outcome ~ arm + baseline)
summary(fit)$coefficients
```

```
# SE comparison
sd(change[arm == "trt"] - mean(change[arm == "trt"]))
```

Output & Results

ANCOVA estimates the treatment effect with substantially smaller SE than the change-score t-test when baseline-outcome correlation is non-trivial.

Interpretation

“ANCOVA estimated the treatment effect as 1.02 (95 % CI 0.74-1.30, $p < 0.001$) with ~40 % lower SE than the change-score analysis, leveraging the 0.7 baseline-outcome correlation.”

Practical Tips

- Use ANCOVA for any continuous outcome where a baseline measurement is available.
- Even with small baseline-outcome correlation (0.3), ANCOVA improves power.
- The “change from baseline” as an outcome is a special case of ANCOVA with $\beta_{\text{base}} = 1$ forced; ANCOVA with free β is preferred.
- Pre-specify baseline adjustment in the SAP; post-hoc addition risks bias.
- EMA guidance on baseline adjustment: stratification and ANCOVA are both acceptable; ANCOVA is more efficient when baseline is continuous.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale